Ameen Ahmed

04/24/2026

ST 533 Project 3 Report

Introduction

This project analyzes crime data in Raleigh, North Carolina using spatial point process methods. The two crime types considered are Larceny and Simple Assault. Each treated as a spatial point pattern.

To study their spatial behavior, we assess complete spatial randomness (CSR) using the L-function and Monte Carlo simulation. We then compare the two-point patterns by estimating a log relative risk surface and visualizing it using a contour plot. A Monte Carlo relabeling approach is used to identify regions where the differences between the two crime types are statistically significant.

Methods

Part 1

We analyze the spatial patterns of Larceny and Simple Assault incidents within the Raleigh township. Each crime type was treated as a spatial point pattern. To reduce computational burden while maintaining spatial structure, a random subset of 100 observations was selected for each crime type. To assess complete spatial randomness (CSR), we use the L-function, which is a transformation of the K-function, and compare the observed results to simulations from a homogeneous Poisson process. Simulation envelopes were generated to determine whether the observed patterns showed clustering, inhibition, or randomness.

Part 2

A log relative risk surface was estimated to compare the spatial distributions between Larceny and Simple Assault. A Monte Carlo relabeling approach was used to assess significance by randomly reassigning crime labels while keeping locations fixed.

Results

Part 1

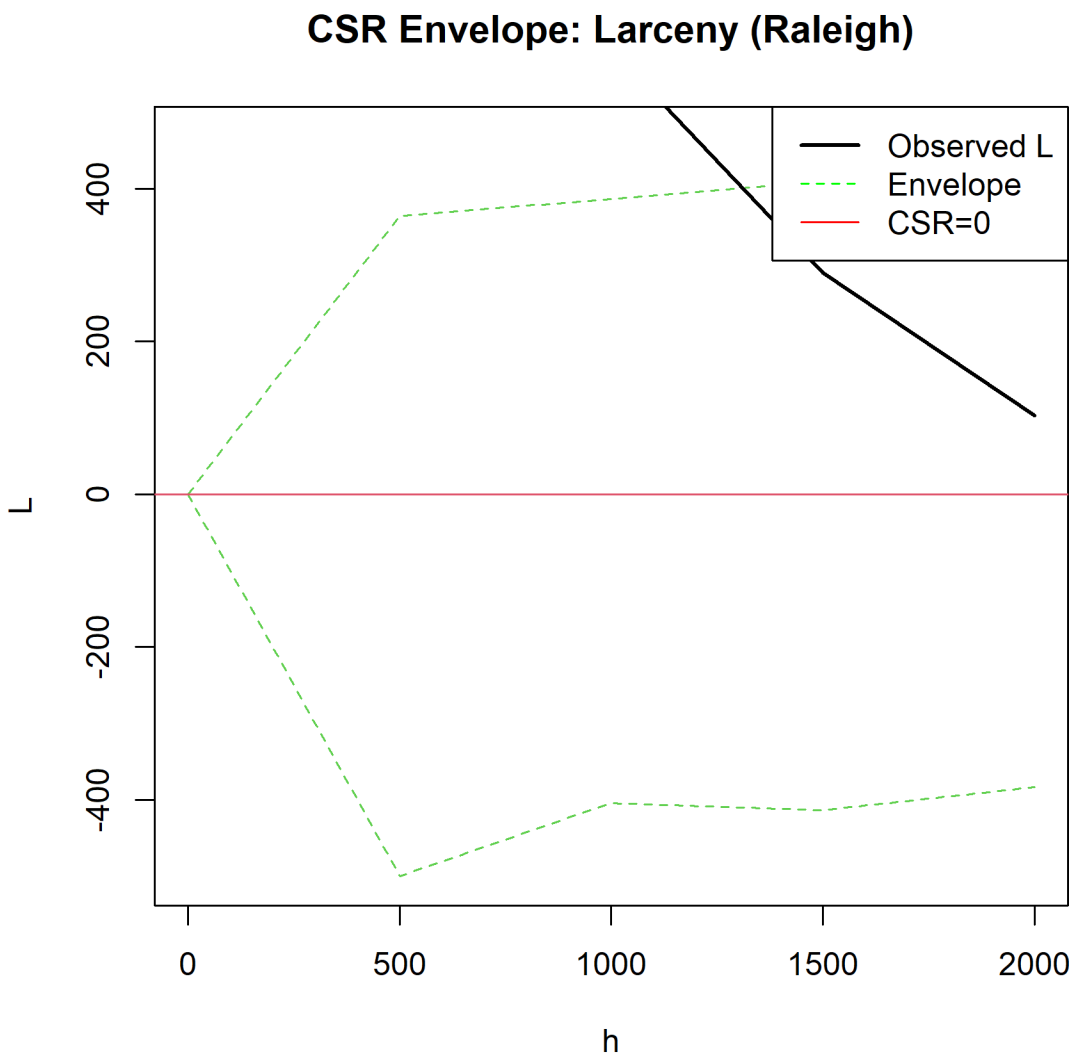

Figure 1L-function with CSR simulation envelope for Larceny incidents in the Raleigh township. The observed L-function (black) is compared to the simulation envelope (green) under complete spatial randomness (CSR = 0 shown in red).

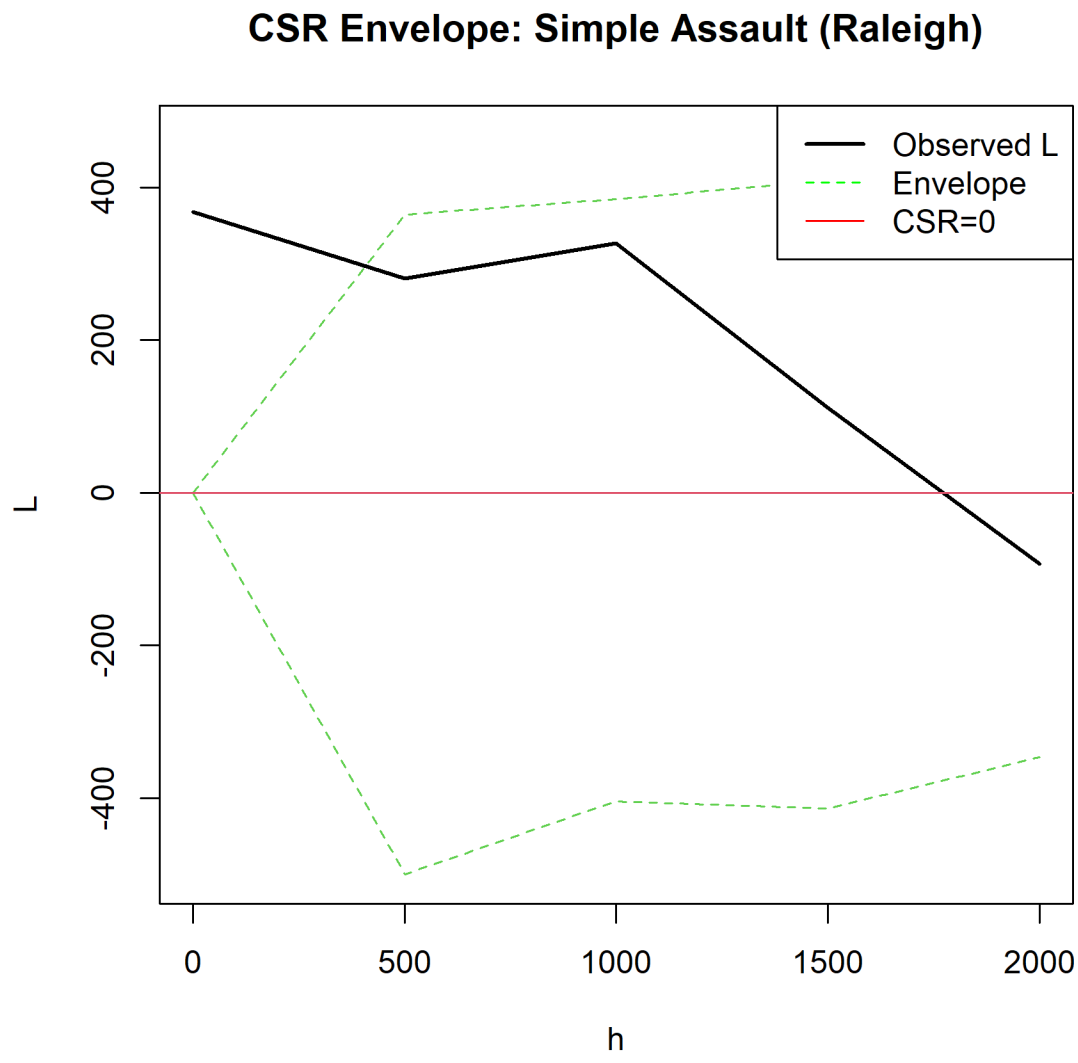

Figure 2CSR envelope for Simple Assault incidents in the Raleigh township. The observed L-function indicates clustering at shorter distances, is consistent with CSR at intermediate distances, and shows inhibition at larger distances.

Interpretation:

The L-function for Larceny lies above the CSR simulation envelope for most distances, indicating strong clustering and a clear deviation from complete spatial randomness.

For Simple Assault, the observed L-function exceeds the envelope at shorter distances, lies within the envelope at intermediate distances, and falls below the envelope at larger

distances. This suggests clustering at shorter distances, randomness at intermediate distances, and inhibition at larger distances.

Part 2

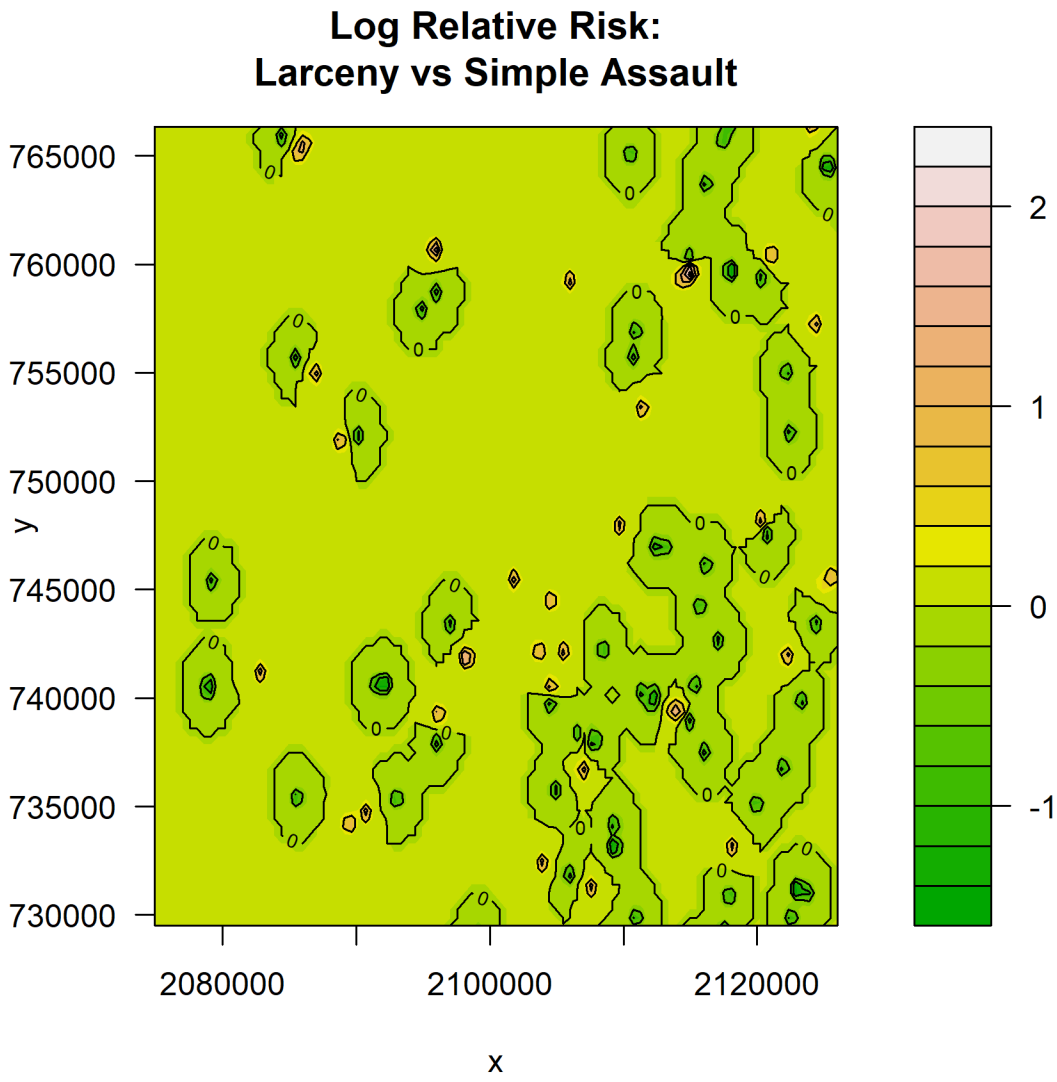

Figure 3Contour map of the log relative risk comparing Larceny and Simple Assault in the Raleigh township. Positive values indicate areas where Larceny is relatively more concentrated, while negative values indicate areas where Simple Assault is relatively more concentrated.

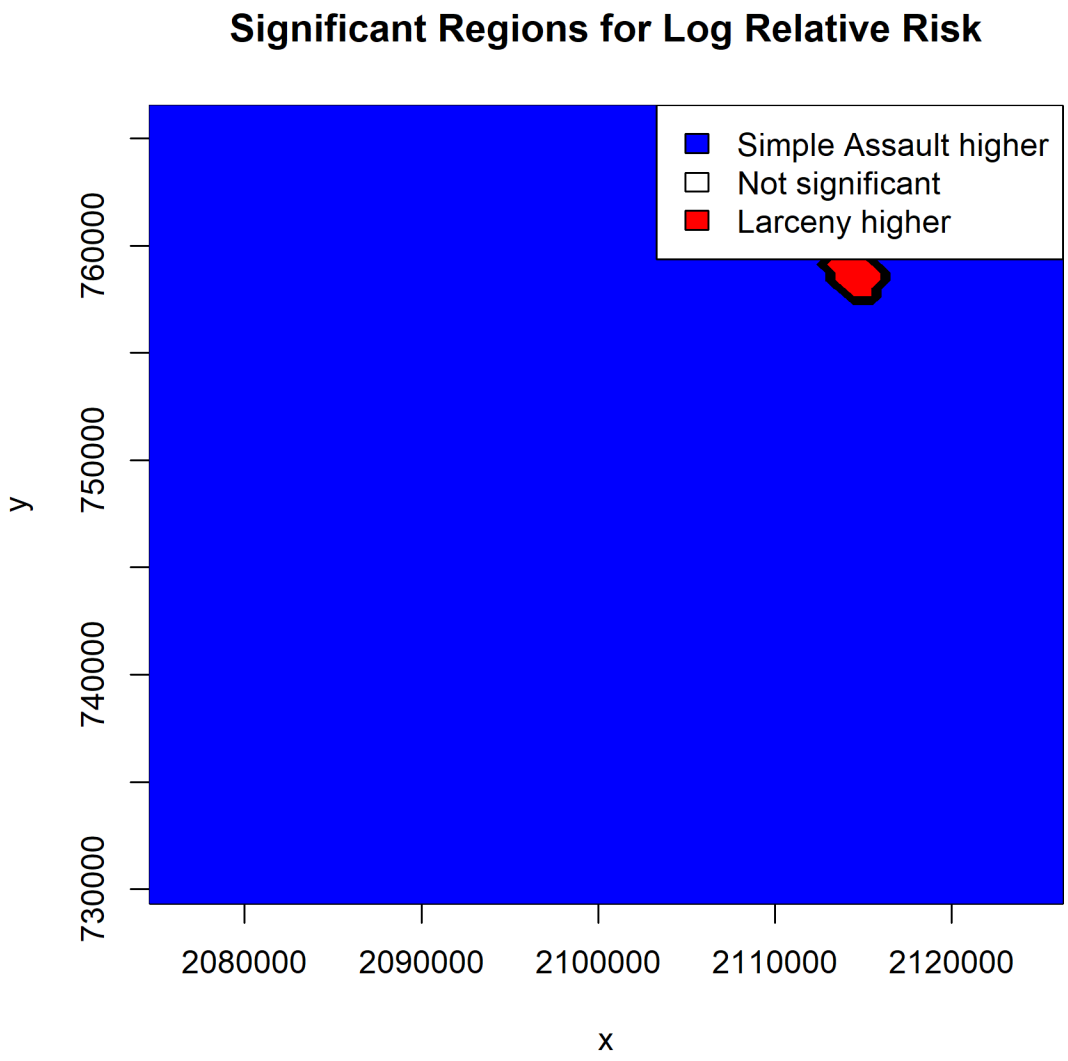

Figure 4Map of statistically significant regions for the log relative risk between Larceny and Simple Assault. Red areas indicate regions where Larceny is significantly more concentrated, blue areas indicate regions where Simple Assault is significantly more concentrated, and white areas indicate no significant difference.

Interpretation:

The log relative risk surface is close to zero across most of Raleigh, indicating that the spatial distributions of Larceny and Simple Assault are generally similar. However, small deviations from zero are present in localized regions.

The significance map shows that many of these small areas are statistically significant, with most areas indicating a higher relative concentration of Simple Assault. Only a small region shows a significantly higher concentration of Larceny.

Discussion

The results show that Larceny exhibits strong clustering, as the L-function lies above the CSR envelope for most distances. In contrast, Simple Assault shows scale-dependent behavior, with clustering at shorter distances, randomness at intermediate distances, and inhibition at larger distances.

The log relative risk surface indicates that the two crime types generally have similar spatial patterns across Raleigh, with most values close to zero. However, the significance map reveals that many of these small distances are statistically meaningful. In particular, Simple Assault is significantly more concentrated across much of the study area, while Larceny is only more concentrated in a small, localized region.

Overall, these results suggests that although the two crime types share similar spatial structures, Simple Assault tends to be relatively more dominant across Raleigh, with Larceny showing stronger clustering but in more limited areas.